

LAB 11

Lab Title: Strings in C++

1. OBJECTIVE

The purpose of this lab is to understand and apply string handling in C++, including C-style strings (character arrays terminated with a null character `\0`), string objects from the `std::string` class, and built-in string manipulation functions such as `strlen()` and `strcat()`.

2. THEORY

A string is a collection of characters. In C++ there are two commonly used types of strings:

- C-strings (C-style strings) — arrays of type `char` terminated with the null character `\0`. The string "Hello" has five characters but occupies six bytes because `\0` is appended automatically. The `cin` extraction operator (`>>`) stops reading at whitespace; `cin.get()` and `getline()` read full lines including spaces.
- String objects — instances of the `std::string` class. Unlike char arrays, string objects have no fixed length and can be extended as required. The `getline()` function is used to read a full line into a string object.

Commonly used built-in string functions for C-strings:

- `strcpy(s1, s2)` — copies string `s2` into string `s1`.
- `strcat(s1, s2)` — concatenates string `s2` onto the end of string `s1`.
- `strlen(s)` — returns the length of string `s`, excluding the null terminator.
- `strcmp(s1, s2)` — returns 0 if the two strings are equal.
- `strlwr(s) /strupr(s)` — converts a string to lowercase or uppercase.

3. SOFTWARE USED

- DEV-C++ IDE
- C++ Compiler (bundled with DEV-C++)

4. PROCEDURE

Task 1 — String Length Calculation

Write a C++ program that declares and initializes a string variable called `message` with a phrase of your choice. Use a built-in function to calculate and display the length of the string.

```
#include <iostream>
#include <cstring>
using namespace std;

int main()
{
    char message[] = "Electrical Engineering"; // initialize string
    int len = strlen(message);                 // calculate length
    cout << "String: " << message << endl;    // display string
    cout << "Length: " << len << endl;        // display length
    return 0;
}
```

Output:

```
String: Electrical Engineering
Length: 25
```

Task 2 — String Concatenation

Declare two string variables called `firstName` and `lastName` and initialize them with your first and last name. Concatenate the two strings to form a full name and display it.

```
#include <iostream>
#include <cstring>
using namespace std;
```

```
int main()
{
    char firstName[50] = "Ab"           // first name
    char lastName[50]  = "ghafar"       // last name (leading
    space)
        strcat(firstName, lastName);    // concatenate strings
    cout << "Full Name: " << firstName << endl; // display result
    return 0;
}
```

Output:

Full Name: ab ghafar

5. CONCLUSION

This lab provided practical experience with string handling in C++. C-style strings were declared and initialized as character arrays. The `strlen()` function was used to determine string length and the `strcat()` function was used to concatenate two strings. The distinction between `cin >>` (which stops at whitespace) and `cin.get()` or `getline()` (which read full lines) was clearly understood. The null terminator `\0` was confirmed as an essential part of every C-style string.